CHAPTER 14

Registers

Storing many bits · shift registers · SISO, SIPO, PISO, PIPO · bidirectional · universal shift register

QUICK TRICK

Read the four letters like this: the first two letters tell how data goes **IN**, the last two how it comes **OUT**. **Serial = one bit at a time (n clock pulses). Parallel = all bits at once (1 clock pulse).**

14.1 What is a register?

● **LEVEL 1 · BEGINNER**

What is it? A flip-flop stores **one bit**. A **register** is a group of flip-flops that stores **many bits** together. An 8-bit register has 8 flip-flops and can store 10110101.

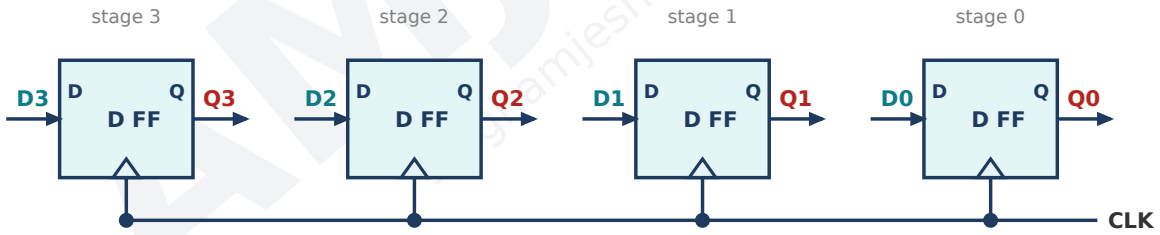
Why do we need it? A computer works on numbers like 8, 16 or 32 bits wide. Registers hold these numbers inside the processor.

Analogy. A row of lockers. Each locker holds one bit. A register is the whole row.

EASY MEMORY

n flip-flops = n-bit register. It can hold 2^n different values. All flip-flops share the **same clock**.

The simplest register (called PIPO, parallel in – parallel out) loads all bits at the same clock pulse:



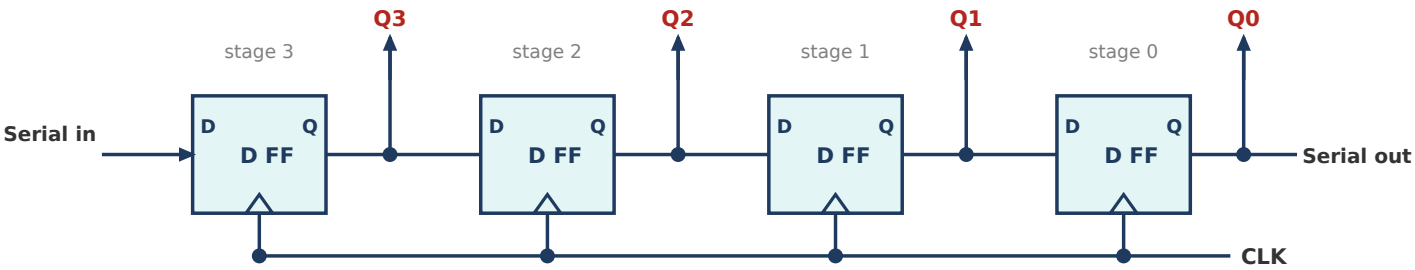
14.2 Shift register

● **LEVEL 1 · BEGINNER**

What is it? A register in which the stored bits **move one place** at every clock pulse. The output of each flip-flop is connected to the input of the next one.

Why? To move data in or out one bit at a time (using only one wire), to delay a signal, and to multiply or divide by 2.

Analogy. A queue of people. When the front person leaves, everyone steps forward and a new person joins at the back.



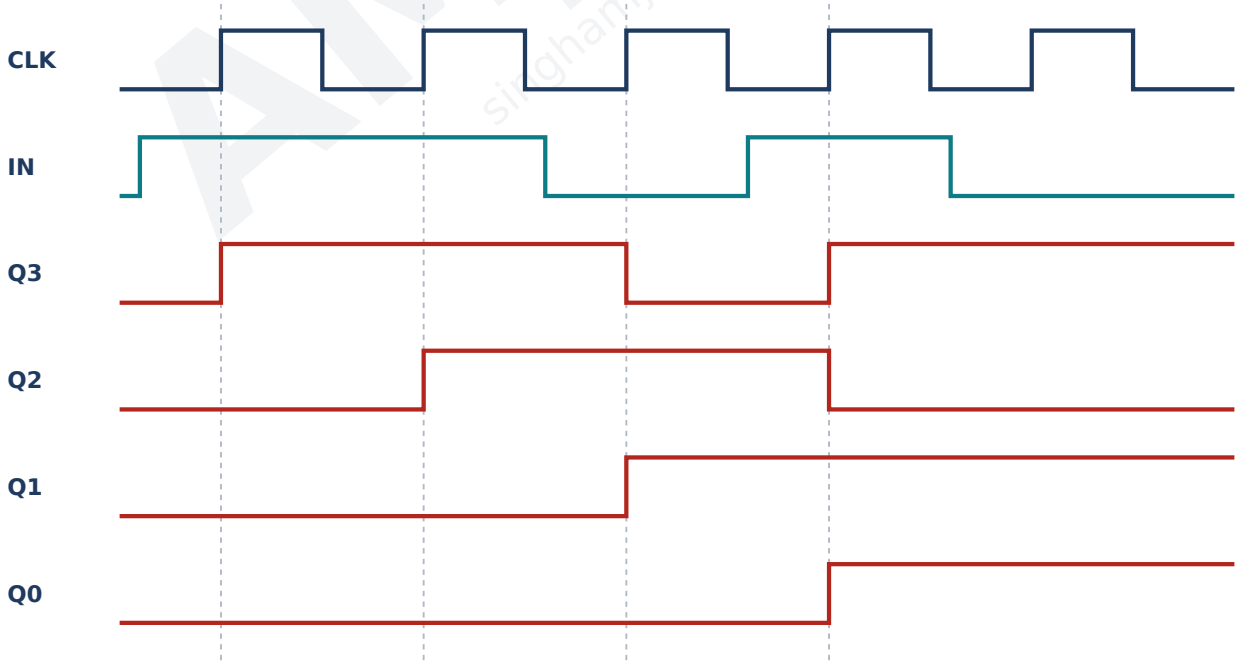
Here data enters at the left and moves to the right. The **first bit that enters travels through all the flip-flops** and reaches the last one.

Example: loading 1011 serially into a 4-bit register

The register starts at 0000. We send the bits **1, 1, 0, 1** (the bit that must end in Q0 goes in **first**).

Clock	Serial in	Q3	Q2	Q1	Q0
0	—	0	0	0	0
1	1	1	0	0	0
2	1	1	1	0	0
3	0	0	1	1	0
4	1	1	0	1	1

After 4 clock pulses the register holds **Q3 Q2 Q1 Q0 = 1011**. Here is the timing diagram. At every rising edge (dashed line) each flip-flop copies the bit on its left, using the value it had **just before** the edge.



QUICK TRICK

Rule for a right-shift register: the first bit entered ends up in the **last** flip-flop (Q0) after n clock pulses. So to get 1011 you send the bits in the order of Q0 first: 1, 1, 0, 1.

14.3 The four types of shift register

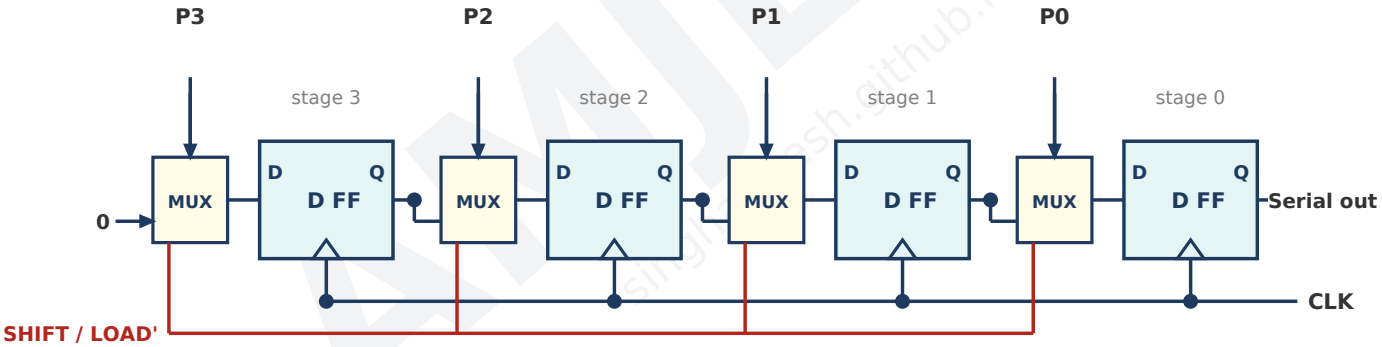
● LEVEL 2 · EXAM

Type	Data goes IN	Data comes OUT	Clock pulses to load	Clock pulses to read	Used for
SISO	Serial	Serial	n	n	Delay line, simple storage
SIPO	Serial	Parallel	n	1 (all bits at once)	Serial → parallel conversion
PISO	Parallel	Serial	1	n	Parallel → serial conversion
PIPO	Parallel	Parallel	1	1	Ordinary storage register

QUICK TRICK
Easy memory: S-I-P-O = Serial In, Parallel Out. Count the clock pulses by looking at the word **serial**: each serial side needs **n** pulses, each parallel side needs **1**.

PISO: parallel in, serial out

Each flip-flop gets a small **2:1 MUX** in front. A control line **SHIFT/LOAD'** chooses: **0 = load** the parallel data P3..P0 (all at once, 1 clock pulse); **1 = shift** (the bit from the left neighbour). After loading, the bits come out one by one at the serial output.



GATE POINT
Serial ↔ parallel conversion: SIPO converts a stream of bits into a parallel word (needs n clocks). PISO does the opposite. Computers use these to talk over a single wire.

14.4 Bidirectional shift register

● LEVEL 2 · EXAM

What is it? A shift register that can shift **right or left**. A control input M chooses the direction. Each flip-flop gets a **2:1 MUX** in front: it can take the bit from its **left** neighbour (shift right) or from its **right** neighbour (shift left).

M	Direction	What happens (Q3 Q2 Q1 Q0)
0	Shift right	Q3 → Q2 → Q1 → Q0. A new bit enters at Q3.
1	Shift left	Q0 → Q1 → Q2 → Q3. A new bit enters at Q0.

QUICK TRICK

Shift = multiply or divide by 2. Shift **left** by 1 = × 2. Shift **right** by 1 = ÷ 2 (the dropped bit is the remainder).
0011 (3) shift left → 0110 (6). 1100 (12) shift right → 0110 (6). Shifting by k places multiplies or divides by 2^k.

Kind of shift	What goes into the empty place	Example (8 bits): 10110011 shifted by 2
Logical shift right	0s	**00101100**
Arithmetic shift right	A copy of the sign bit (keeps the sign of a 2's complement number)	**11101100**
Rotate (circular) left by 1	The bit that fell off the other end	10010110 → **00101101**

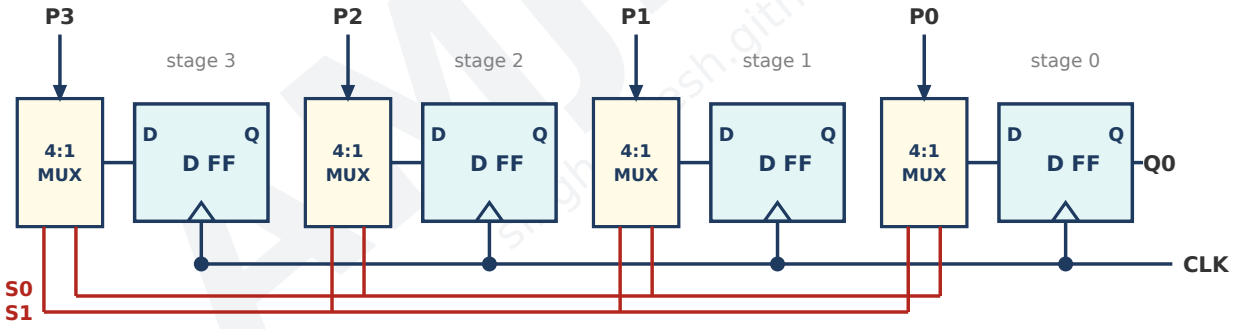
EXAM TRAP

Arithmetic shift right copies the sign bit, but a **logical shift** puts 0s. They give different answers for negative numbers. A **rotate** loses no bit: it joins the output back to the input.

14.5 Universal shift register

● **LEVEL 3 · GATE**

What is it? One register that can do **everything**: hold, shift right, shift left and parallel load. (The IC 74194 is a 4-bit universal shift register.) Each flip-flop gets a **4:1 MUX**; two select lines S1 S0 choose the mode.



S1	S0	Mode	What each flip-flop takes as input
0	0	**Hold**	Its own output (no change)
0	1	**Shift right**	The output of its left neighbour (serial-right input enters at Q3)
1	0	**Shift left**	The output of its right neighbour (serial-left input enters at Q0)
1	1	**Parallel load**	The parallel input P

QUICK TRICK

Mode code: Hold, Right, Left, Load = 0, 1, 2, 3. Read S1 S0 as a binary number. 00 = 0 = Hold. 01 = 1 = shift Right. 10 = 2 = shift Left. 11 = 3 = Load. An n-bit universal register needs **n 4:1 MUXes**.

Example run (Q3 Q2 Q1 Q0):

Step	S1 S0	Mode	Serial/parallel inputs	Q3 Q2 Q1 Q0
start	—	—	—	0000
1	1 1	Load	P3..P0 = 1010	1010
2	0 1	Shift right	serial in (right) = 1	1101
3	1 0	Shift left	serial in (left) = 0	1010
4	0 0	Hold	—	1010

14.6 A shift register can also count (preview of Chapter 15)

● **LEVEL 3 · GATE**

If the **last output is fed back to the first input**, the bits go round and round. This makes counters:

Counter	Feedback	Sequence for 4 bits	States
Ring counter	Q0 → serial input	1000 → 0100 → 0010 → 0001 → repeat (start 1000)	n
Johnson counter	Q0' → serial input	0000 → 1000 → 1100 → 1110 → 1111 → 0111 → 0011 → 0001 → repeat	2n

GATE POINT

n flip-flops: **ring counter = n states**, **Johnson counter = 2n states** (the inverted feedback doubles the count). (Chapter 15 explains them in detail.)

Quick-Solve Tricks — Summary

Topic	Trick
Bits stored	n flip-flops = n-bit register.
Serial vs parallel	Serial = n clock pulses. Parallel = 1 clock pulse.
First bit in	Ends in the last flip-flop after n clock pulses.
Shift = × 2 or ÷ 2	Left × 2, right ÷ 2 (k places = 2^k).
Universal register	S1 S0 = 00 Hold, 01 Right, 10 Left, 11 Load. n 4:1 MUXes.
Ring / Johnson	Ring = n states, Johnson = 2n states.

GATE QUICK CHECK — Chapter 14

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n-bit register = n flip-flops          Serial = n clocks, Parallel = 1 clock
Shift left = x 2, shift right = / 2
Delay of an n-bit shift register = n clock periods
Universal register S1 S0 : 00 hold, 01 right, 10 left, 11 load
Ring counter n states, Johnson counter 2n states
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Traps: order of the bits when loading serially · PISO needs a MUX and a load control · arithmetic vs logical shift · Johnson counter has 2n states (not n).

Practice questions

● LEVEL 1 · BEGINNER

Q1. How many flip-flops are needed for an 8-bit register?

Q2. What does PISO stand for?

Q3. What is the effect of shifting 0011 left by one place?

● LEVEL 2 · EXAM

Q4. A 4-bit right-shift register starts at 0000. The serial input is 1, 0, 1, 1 (in this order), entering at Q3. Find Q3 Q2 Q1 Q0 after 4 clock pulses.

Q5. A 10 MHz clock drives an 8-bit SISO register. How long does it take to load 8 bits serially?

Q6. A 4-bit universal shift register holds 0110. S1 S0 = 10 (shift left) and the serial-left input is 1. Find the new value.

● LEVEL 3 · GATE

Q7. A 4-bit ring counter starts at 1000. Find the value after 3 clock pulses.

Q8. A 4-bit Johnson counter starts at 0000. Find the value after 3 clock pulses.

Q9. An 8-bit register holds 10110011. Find the result of (a) a logical shift right by 2 and (b) an arithmetic shift right by 2.

Q10. Rotate 10010110 left by one place.

Q11. How many 4:1 MUXes does a 4-bit universal shift register need?

Q12. A 4-bit PISO register (right shift, serial output at Q0) is loaded in parallel with Q3 Q2 Q1 Q0 = 1011. In what order do the bits come out?

Answers

Q	Solution
1	**8** flip-flops.
2	**P**arallel **I**n, **S**erial **O**ut.
3	Shift left = × 2: 0011 (3) → **0110** (6).
4	After clock 1: 1000. Clock 2 (input 0): 0100. Clock 3 (input 1): 1010. Clock 4 (input 1): **1101**.
5	Clock period = 1 / 10 MHz = 100 ns. 8 pulses × 100 ns = **800 ns**.
6	Shift left: Q3 ← Q2 = 1, Q2 ← Q1 = 1, Q1 ← Q0 = 0, Q0 ← serial-left = 1. New value **1101** (= 6 × 2 + 1 = 13).
7	1000 → 0100 → 0010 → **0001**.
8	0000 → 1000 → 1100 → **1110**.
9	(a) Logical: **00101100**. (b) Arithmetic (sign bit 1 is copied): **11101100**.
10	The leftmost 1 goes to the right end: 1001 0110 → **0010 1101**.
11	One 4:1 MUX per flip-flop → **4**.
12	Q0 comes out first: **1, 1, 0, 1** (Q0 = 1, Q1 = 1, Q2 = 0, Q3 = 1). The first bit is available right after loading; the remaining 3 need 3 more clock pulses.

Next: Chapter 15 — Counters (ripple/asynchronous, synchronous, up/down, mod-N, decade, ring, Johnson, frequency division, design).